Course PH410 Course coordinator: Alika Khare

Problem Set _09 Due Date: April 24-25, 2025

1. The electric field of an EM wave in free space is given by $\vec{E} = 50\hat{i}\cos(10^8 t + kz)$ V/m , find the time taken by the wave to travel a distance of $\lambda/2$. Sketch the electric field as a function of z for t=0, $T/4$ and $T/2$, where T is the time period of the wave.

2. An electromagnetic wave of frequency ω and having magnitude of propagation vector k is propagating along x axis in air. The wave is polarized along z axis. Write down the expression for the electric field and magnetic field.

3. The electric field in a non magnetic medium is given by $\boldsymbol{E} = 50\,cos(10^9 t - 8x)\hat{j}$+40sin$(10^9 t - 8x)\hat{k}$. Find the expression for corresponding $\boldsymbol{H}$.

4. The expression for $\boldsymbol{H}$ (magnetic field intensity) associated with the radiation in free space from an antenna is given by $\vec{H} = \frac{12\sin\theta}{r}\cos(2\pi x 10^8 t - kr)\hat{\theta}\ mA/m$. Find the electric field using one of the suitable Maxwell's equation and Poynting vector.

5. The electric field of an spherical wave is given by $\vec{E} = \frac{E_0}{r} sin\theta Cos\big(\omega(t - r/c)\big)\hat{\theta}$. Plot E as a function of r for few cycles. Obtain the expression for magnetic field and poynting vector.

6. The dispersion relation for a particular medium is given by $k^2 = \omega^2 A\left(1 - \frac{B}{\omega^2}\right)$, where A and B are constant. Find the group velocity for this medium. In what range of ω, wave will not propagate through the medium.

7. Prove equation nos. 9.111 and hence 9.112 from Griffith (4th edition) page no.410

8. Problems from Griffith fourth edition: 9.18 and 9.25.

1. The electric field of an EM wave in free space is given by $\vec{E} = 50\hat{i}\cos(10^8 t + kz)$ V/m, find the time taken by the wave to travel a distance of $\lambda/2$. Sketch the electric field as a function of z for t=0, T/4 and T/2, where T is the time period of the wave.

$$\vec{E} = 50\hat{i}\cos(10^8 t + kz)$$

$$\vec{E} = 50\hat{i}\cos(-10^8 t - kz)$$

$$\vec{E} = 50\hat{i}\cos(-kz - 10^8 t)$$

$$\omega = 10^8$$

$$c = \lambda f$$

$$c = \frac{2\pi}{2\pi}\lambda f$$

$$c = \frac{\omega}{k}$$

$$k = \frac{\omega}{c} = \frac{10^8}{3\times10^8} = \frac{1}{3}\ \text{rad/m}$$

$$\lambda/2 = \frac{2\pi}{k}\times\frac{1}{2} = \frac{2\pi\times 3}{2} = 3\pi\ \text{m}$$

$$t = \frac{\lambda/2}{c} = \frac{3\pi}{3\times10^8} = \pi\times10^{-8}\ \text{s}$$

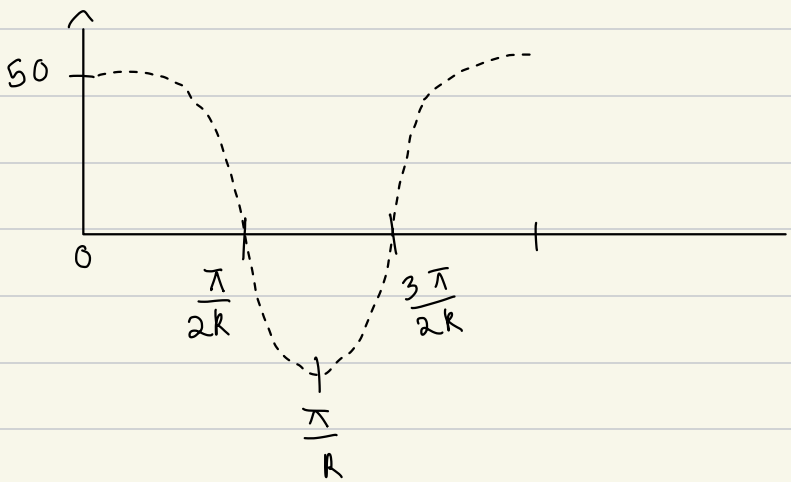
$$\vec{E} = 50\hat{i}\cos(-kz - 10^8 t)$$

$t = 0$,

$$\vec{E} = 50\,\hat{i}\cos(kz)$$

$\cos(kz) = -1$

$kz = (2n+1)\pi$



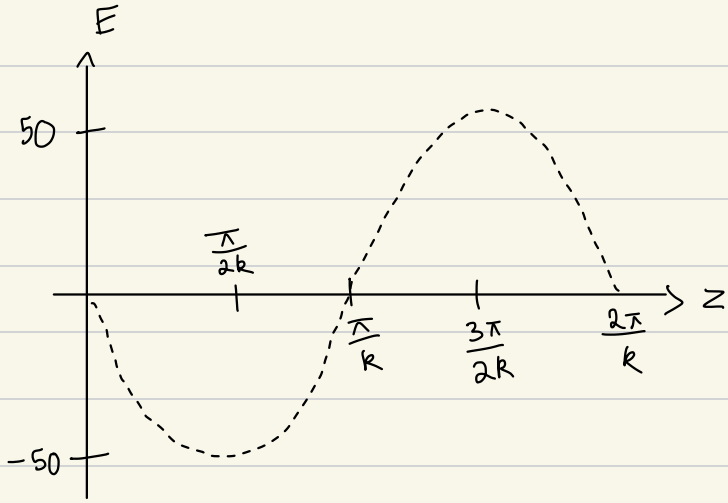
$t = T/4$

$$\frac{T}{4} = \frac{2\pi}{\omega}\frac{1}{4} = \frac{\pi}{2\omega} = \frac{\pi}{2}10^{-8}$$

$$\vec{E} = 50\hat{i}\cos\left(-kz - 10^8\frac{\pi}{2}10^{-8}\right)$$

$$= 50\hat{i}\cos\left(\frac{\pi}{2} + kz\right)$$

$$= -50\hat{i}\sin(kz)$$



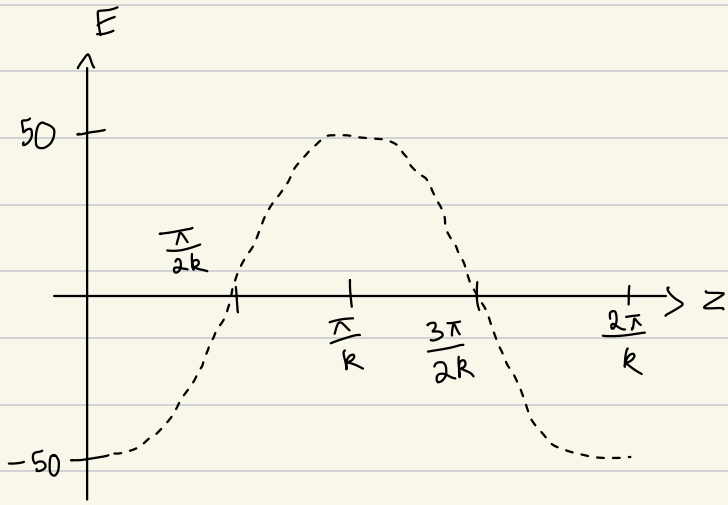
$t = T/2$

$$\frac{T}{2} = \frac{2\pi}{\omega}\,\frac{1}{2} = \frac{\pi}{2\omega} = \pi \cdot 10^{-8}$$

$$\vec{E} = 50\hat{i}\cos\left(-kz - 10^{8}\,\pi\,10^{-8}\right)$$

$$= 50\hat{i}\cos\left(\pi + kz\right)$$

$$= -50\hat{i}\cos\left(kz\right)$$



2. An electromagnetic wave of frequency ω and having magnitude of propagation vector k is propagating along x axis in air. The wave is polarized along z axis. Write down the expression for the electric field and magnetic field.

$$\vec{E} = E_0 \hat{z} e^{i(kx-\omega t)}$$

$$\vec{B} = \frac{1}{\omega} \vec{k} \times \vec{E}$$

$$\vec{B} = -\frac{E_0}{c} \hat{y} e^{i(kx-\omega t)}$$

$$= \frac{k}{\omega} E_0 (-\hat{y}) = \frac{E_0}{c} (-\hat{y})$$

3. The electric field in a non magnetic medium is given by $\boldsymbol{E} = 50\, cos(10^9 t - 8x)\hat{j} + 40\sin(10^9 t - 8x)\hat{k}$. Find the expression for corresponding $\boldsymbol{H}$.

$$\vec{E} = 50 \cos(10^9 t - 8x)\hat{j} + 40 \sin(10^9 t - 8x)\hat{k}$$

$$\vec{k} = k\hat{x} = 8\hat{x}$$

$$\vec{B} = \frac{1}{\omega} \vec{k} \times \vec{E}$$

$$\vec{B} = \frac{8}{10^9} \left\{ 50 \cos(10^9 t - 8x)\hat{z} - 40 \sin(10^9 t - 8x)\hat{y} \right\}$$

$$\vec{H} = \frac{\vec{B}}{\mu_0} = \frac{8 \times 10^{-8}}{\mu_0} \left\{ 5 \cos(10^9 t - 8x)\hat{z} - 4 \sin(10^9 t - 8x)\hat{y} \right\}$$

4. The expression for $\boldsymbol{H}$ (magnetic field intensity) associated with the radiation in free space from an antenna is given by $\vec{H} = \frac{12\sin\theta}{r}\cos(2\pi x 10^8 t - kr)\hat{\theta}\ mA/m$. Find the electric field using one of the suitable Maxwell's equation and Poynting vector.

$$\vec{H} = \frac{12\sin\theta}{r}\cos(2\pi \times 10^8 t - kr)\,\hat{\theta}$$

$$\vec{\nabla}\times\vec{H} = \frac{1}{r^2\sin\theta}\begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\,\hat{\varphi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\varphi} \\ H_r & rH_\theta & r\sin\theta\, H_\varphi \end{vmatrix}$$

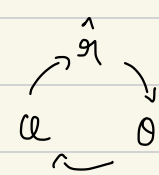
$$= \frac{1}{r^2\sin\theta}\begin{vmatrix} \hat{r} & r\hat{\theta} & r\sin\theta\,\hat{\varphi} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\varphi} \\ 0 & rH_\theta & 0 \end{vmatrix}$$

$$= \frac{1}{r}\hat{\varphi}\left\{\frac{\partial}{\partial r}\,12\sin\theta\cos(2\pi\times 10^8 t - kr)\right\}$$

$$= -\frac{12}{r}\sin\theta\, k\sin(2\pi\times 10^8 t - kr)\,\hat{\varphi}$$

$$D = -\frac{12}{r}\sin\theta\, k\int\sin(2\pi\times 10^8 t - kr)\,dt\,\hat{\varphi}$$

$$\vec{E} = \frac{12\sin\theta\, k\cos(2\pi\times 10^8 t - kr)}{2\pi\varepsilon\times 10^8\, r}\,\hat{\varphi}$$



$$\vec{S} = \vec{E}\times\vec{H} = \left(\frac{12\sin\theta}{r}\right)^2\left(\frac{k}{2\pi\varepsilon\times 10^8}\right)\cos^2(2\pi\times 10^8 t - kr)\,(-\hat{r})$$

5. The electric field of an spherical wave is given by $\vec{E} = \frac{E_0}{r} sin\theta Cos\big(\omega(t - r/c)\big)\hat{\theta}$. Plot E as a function of r for few cycles. Obtain the expression for magnetic field and poynting vector.

$$\vec{E} = \frac{E_0}{r} \sin\theta \cos(\omega t - kr)\,\hat{\theta}$$

$$\vec{k} = k\,\hat{r}$$

$$\vec{B} = \frac{1}{\omega}(\vec{k} \times \vec{E})$$

$$\vec{B} = \frac{E_0}{rc} \sin\theta \cos(\omega t - kr)\,\hat{\varphi}$$

$$\vec{S} = \left(\frac{E_0}{r}\right)^2 \frac{1}{c\mu_0} \sin^2\theta \cos^2(\omega t - kr)\,\hat{r}$$

6. The dispersion relation for a particular medium is given by $k^2 = \omega^2 A\left(1 - \frac{B}{\omega^2}\right)$, where A and B are constant. Find the group velocity for this medium. In what range of ω, wave will not propagate through the medium.

$$k^2 = \omega^2 A\left(1 - \frac{B}{\omega^2}\right) \Rightarrow k = \sqrt{A(\omega^2 - B)}$$

$$2k\frac{dk}{d\omega} = A\,2\omega\left(1 - \frac{B}{\omega^2}\right) + A\omega^2\,2\frac{B}{\omega^3}$$

$$\frac{d\omega}{dk} = \left[A\frac{\omega}{k}\left(1 - \frac{B}{\omega^2}\right) + \frac{AB}{\omega k}\right]^{-1}$$

$$= \left[\frac{A\omega^3 - A\omega B + A\omega B}{k\omega^2}\right]^{-1}$$

$$= \frac{k}{A\omega} = \frac{[A(\omega^2 - B)]^{1/2}}{A\omega}$$

9.18

$n_2 = 2.42$

$$\alpha = \frac{\sqrt{1-\sin^2\theta_T}}{\cos\theta_I} = \frac{\sqrt{1-[(n_1/n_2)\sin\theta_I]^2}}{\cos\theta_I}$$

$$\beta = \frac{\mu_1 n_2}{\mu_2 n_1} = \frac{n_2}{n_1} = 2.42 \quad [\because \mu_1 = \mu_2 = \mu_0]$$

$$\alpha = \frac{\sqrt{1-[\sin\theta_I/2.42]^2}}{\cos\theta_I}$$

i)

$\theta_I = 0 \Rightarrow \alpha = 1$ and $\beta = 2.42$

$$\frac{E_{0R}}{E_{0I}} = \frac{\alpha-\beta}{\alpha+\beta} = \frac{1-2.42}{1+2.42} = -0.415$$

$$\frac{E_{0T}}{E_{0I}} = \frac{2}{\alpha+\beta} = \frac{2}{3.42} = 0.585$$

ii) θ for which $E_{0R} = 0$

$$\theta_B = \tan^{-1}\left(\frac{n_2}{n_1}\right) = 67.5^\circ$$

iii) θ at which $E_{0R} = E_{0T}$

$$\frac{E_{0R}}{E_{0I}} = \frac{E_{0T}}{E_{0I}} \Rightarrow \frac{\alpha - \beta}{\alpha + \beta} = \frac{2}{\alpha + \beta} \Rightarrow \alpha = 2 + \beta = 4.42$$

$$\alpha = \frac{\sqrt{1 - [\sin\theta_I / 2.42]^2}}{\cos\theta_I}$$

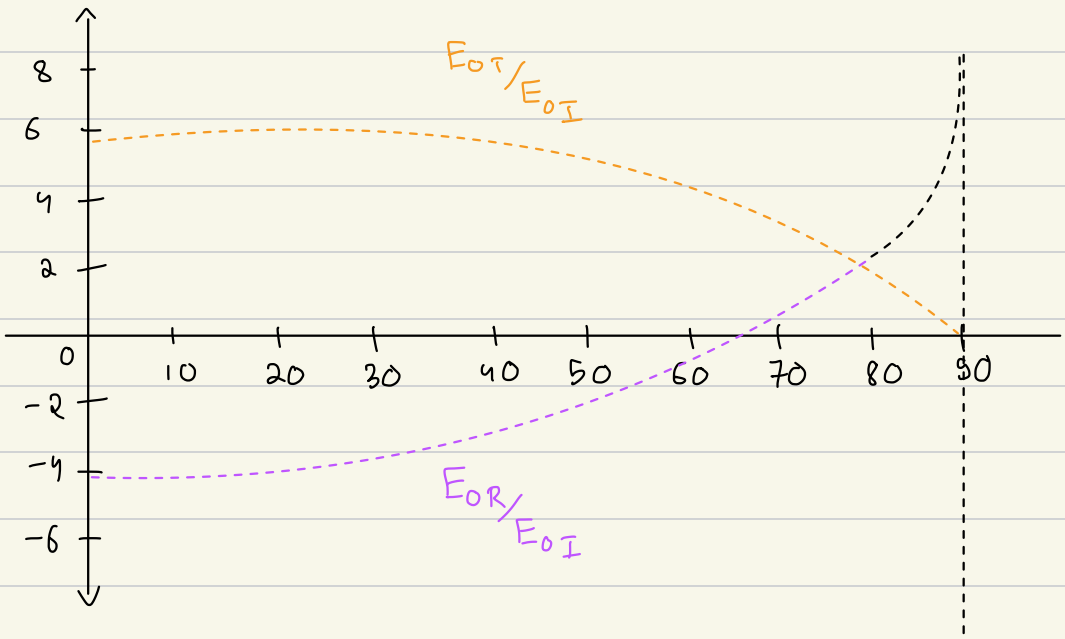
$$(4.42)^2 \cos^2\theta_I = 1 - \frac{\sin^2\theta_I}{(2.42)^2}$$

$$(4.42)^2 - (4.42)^2 \sin^2\theta_I = 1 - \frac{\sin^2\theta_I}{(2.42)^2}$$

$$(4.42)^2 - 1 = \sin^2\theta_I \left[(4.42)^2 - \frac{1}{(2.42)^2} \right]$$

$$\sin^2\theta_I = 0.9572$$

$$\theta_I = 78.06$$


8
6
4
2
0
-2
-4
-6
10
20
30
40
50
60
70
80
90
E_{0T}/E_{0I}
E_{0R}/E_{0I}

9.25